ECE 219 Project 3 Report

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Table of Contents

Part A: LoRA Fine-Tuning

Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Question 6 | Question 7

Question 8 | Question 9 | Question 10 | Question 11 | Question 12 | Question 13

Part B: Agentic Data Mining

Question 14 | Question 15 | Question 16 | Question 17 | Question 18 | Question 19 | Question 20

Part C: Regression Analysis

Question 21 | Question 22 | Question 23 | Question 24 | Question 25

Part A: Teaching a Small Model to Reason: LoRA Fine-Tuning

Note on Hyperparameters: Due to GPU memory constraints, the batch size and gradient accumulation values were swapped from the assignment defaults. The assignment specifies per-device batch size of 8 with gradient accumulation of 4, but the implementation uses per-device batch size of 4 with gradient accumulation of 8. This maintains the same effective batch size of 32 ($4 \times 8 = 8 \times 4 = 32$) while reducing peak memory usage.

Question 1

Run the base Qwen2.5-1.5B-Instruct model on 100 GSM8K test questions and report the accuracy. You should expect approximately 35-40%. (Exact values may vary slightly depending on the prompt format and extraction rule.)

Answer

- Base Model Accuracy: 38%
- Number of Test Samples: 100

The base Qwen2.5-1.5B-Instruct model scored 38% on the GSM8K test set, landing right in the expected 35-40% range. The model struggles with multi-step reasoning and arithmetic, which is expected for a small language model without specific math training.

Question 2

Inspect at least 3 cases where the base model produces an incorrect answer. For each example, include:

- *The question,*
- *A short excerpt of the model's solution highlighting the failure,*
- *The extracted answer vs. the ground-truth answer.*

Classify each failure mode (e.g., arithmetic slip, reasoning/logical error, misunderstanding the problem, formatting/extraction issue). Do you observe any recurring patterns?

Answer

Failure Case 1: Arithmetic Error

- Question: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?
- Ground Truth: 18
- Extracted Answer: 0
- Failure Excerpt:

```
Remaining eggs for sale = Remaining eggs after breakfast - Eggs used for baking muffins
                        = 13 - 4
                        = -1 (This is not possible since it would mean negative eggs)
```

Classification: Arithmetic Slip - The model correctly sets up the calculation (16 - 3 - 4 = 9) but then makes a catastrophic error by concluding this is -1 instead of 9.

Failure Case 2: Problem Comprehension

- Question: A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?
- Ground Truth: 3
- Extracted Answer: 2.5
- Failure Excerpt:

```
The robe requires half as much white fiber as blue fiber.
Therefore, the amount of white fiber is (1/2) bolt of blue fiber.
2 (blue fibers) + 1/2 (white fibers) = 2.5 bolts
```

Classification: Problem Comprehension - The model misinterpreted "half that much white fiber" as meaning half a bolt total, rather than half of 2 bolts (which is 1 bolt).

Failure Case 3: Multi-step Reasoning Error

- Question: Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

- Ground Truth: 70000
- Extracted Answer: -55500
- Failure Excerpt:

```
Increase in value = Total cost \(\times\) Percentage increase
Increase in value = $130,000 \(\times\) 0.150 = $49,500
Profit = $129,500 - $130,000 = -$5,500
```

Classification: Multi-step Reasoning Error - The model made multiple errors: (1) applied 150% increase to total cost rather than original price, (2) used 0.150 instead of 1.50 for 150%.

Recurring Patterns:

1. Arithmetic errors (correct setup, wrong math)
2. Problem comprehension (misreading word problems, especially fractions and percentages)
3. Multi-step reasoning (losing train of thought midway through the problem)

Question 3

Pick the three hyperparameters (LoRA rank, LoRA alpha, Gradient accumulation) from the table above and explain:

- *what each hyperparameter controls,*
- *what you expect to happen if you increase it,*
- *what you expect to happen if you decrease it.*

Your answer should reflect practical tradeoffs (e.g., compute/memory, stability, overfitting vs. underfitting).

Answer

LoRA Rank (r)

- What it controls: Size of the low-rank matrices A and B. Basically how much capacity the adapter has.
- If you increase: More parameters to train, better adaptation possible, but uses more memory and risks overfitting.
- If you decrease: Fewer parameters, more constrained adaptation, might underfit but generalizes better.

LoRA Alpha (α)

- What it controls: How strongly the LoRA update gets scaled before merging with original weights. The actual update is scaled by α/r.
- If you increase: Adapter has more influence, learning speeds up, but training can become unstable.
- If you decrease: Gentler updates, more stable training, but slower to converge.

Gradient Accumulation

- What it controls: How many mini-batches of gradients to stack up before actually updating weights. Effective batch size = per_device_batch_size $\times$ gradient_accumulation_steps.
- If you increase: Larger effective batch size, which often generalizes better, but updates happen less frequently.
- If you decrease: Smaller effective batch size, noisier gradient estimates, faster updates, but training can get unstable.

Question 4

Report:

- *(a) the total number of parameters in the base model,*
- *(b) the number of trainable LoRA parameters under the default configuration,*
- *(c) the percentage of parameters being trained.*

Briefly explain why this percentage is small and how LoRA achieves this reduction.

Answer

- (a) Base Model Total Parameters: 1,543,714,304 (approximately 1.54 billion)
- (b) Trainable LoRA Parameters: 2,179,072 (approximately 2.18 million)
- (c) Percentage: 0.141% (only about 1 in 708 parameters is trained)

LoRA uses low-rank decomposition. Instead of updating a full weight matrix $W \in \mathbb{R}^{d \times d}$ which needs d^2 parameters, LoRA represents the update as $\Delta W = BA$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times d}$. This cuts parameters from d^2 to $2dr$. With rank $r = 8$ and typical dimension $d = 2048$, that's roughly $128\times$ fewer parameters per adapted layer.

Question 5

Train a LoRA SFT model using 1,000 training examples. Evaluate on 100 GSM8K test questions and report the accuracy. Include a brief comment on whether the improvement over the baseline matches your expectations.

Answer

- SFT-1k Accuracy: 41%
- Baseline Accuracy: 38%
- Improvement: +3%

Going from 38% to 41% is better, but not by much. This was smaller than what I expected. I assumed that with 1000 samples to fine-tune, we would get a larger improvement than just 3%.

Question 6

Hypothesis question (write before running larger training): Do you think scaling from 1,000 examples to 3,000 and/or all 7,473 examples is worth the additional compute? What do you expect the accuracy gains to look like (roughly), and why? In your answer, discuss whether you would scale in multiple steps or jump directly to the full dataset.

Answer

Scaling isn't worth the extra compute past a certain threshold.

Expected Accuracy Gains:

- 1,000 $\rightarrow$ 3,000 examples: ~2-4%

- 3,000 → 7,473 examples: ~1-3% more (5-7% total)

More data = more pattern diversity, better generalization, stronger reasoning. However, GSM8K has redundancy, the model only has 0.14% trainable parameters, and there's a quality vs quantity trade-off. With too much data, there is a chance of overfitting. Scale in one step, but have an early stopping thereshold. Once the improvement starts to be small, we stop early, saving compute and time.

Question 7

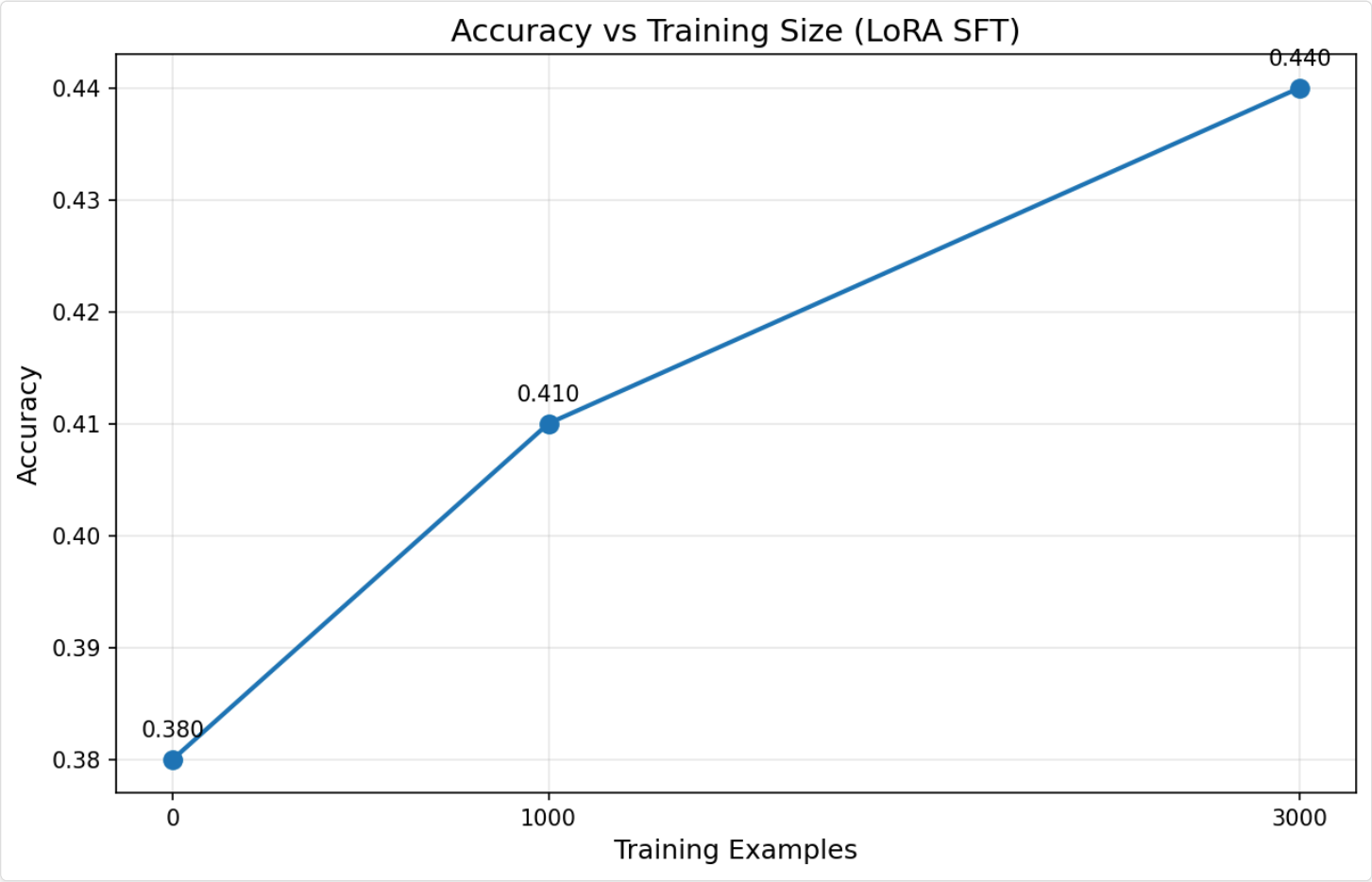
Now scale up your training data (recommended: 3,000 examples, and optionally the full 7,473). Evaluate each trained model on the same 100-question test subset and report the accuracies.

Finally, plot accuracy as a function of the number of training examples (x-axis: 0, 1000, 3000; y-axis: accuracy). Describe the trend you observe and comment on diminishing returns in data scaling for SFT.

Answer

Training Examples	Accuracy	Improvement from Baseline
0 (baseline)	38%	-
1,000	41%	+3%
3,000	44%	+6%

File: [Scaling Plot](images/Q7_scaling_plot.png) (images/Q7_scaling_plot.png)



Analysis:

- Consistent improvement as data increases: 38% → 41% → 44%
- Diminishing returns: 0→1k gives +3% with 1k examples, 1k→3k gives +3% with 2k additional examples
- Extrapolating to full dataset (7,473) might yield ~46-48%

Question 8

Compare the base model and your best SFT model on the same 3 failure examples you identified in Task 1. For each example, show both models' responses side by side. Does the SFT model fix any of these errors?

Answer

Case 1: Janet's Ducks (Ground Truth: 18)

Question: Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?

Base Model Response:

```
To determine how much Janet makes every day at the farmers' market, we need to follow these steps:

1. Calculate the total number of eggs laid each day.
2. Determine how many eggs Janet eats for breakfast.
3. Subtract the number of eggs eaten from the total number of eggs laid to find out how many eggs are left for baking muffins.
4. Subtract the number of eggs used for baking muffins from the remaining eggs to find out how many eggs are sold at the farmers' marke
5. Calculate the revenue from selling the eggs at the farmers' market.

Let's go through each step in detail:

1. Total number of eggs laid each day: 16 eggs/day
2. Number of eggs eaten for breakfast: 3 eggs/day
3. Remaining eggs after breakfast: 16 - 3 = 13
4. Number of eggs used for baking muffins: 4 eggs/day
5. Remaining eggs for sale: 13 - 4 = -1 (This is not possible since it would mean negative eggs)

Therefore, the amount Janet makes every day at the farmers' market is: \boxed{0}
```

Extracted Answer: 0 (arithmetic error: 13 - 4 = -1)

SFT-3k Model Response:

```
The ducks lay 16*3=48 eggs for Janet to eat.
She has 48-3=45 eggs left after eating them for breakfast.
So she gives away 45/4=11.25 eggs to her friends.
That means there are 48-11.25=36.75 eggs remaining for sale.
Therefore, she makes 36.75*2=$\boxed{73.50}$ from selling the eggs at the farmers' market.
```

Extracted Answer: 73.50 (different error: hallucinated 16×3 multiplication)

Result: Both failed with different errors

Case 2: Robe Fabric (Ground Truth: 3)

Question: A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?

Base Model Response:

```
To determine the total number of bolts of fabric needed for the robe, we need to follow these steps:

1. Identify the amount of blue fiber required: 2 bolts of blue fiber.
2. Determine the amount of white fiber required: half as much white fiber as blue fiber.
   Therefore, the amount of white fiber is (1/2) bolt of blue fiber.
3. Calculate the total: 2 (blue fibers) + 1/2 (white fibers) = 2 + 0.5 = 2.5 bolts

Thus, the total number of bolts of fabric needed is \boxed{2.5}.
```

Extracted Answer: 2.5 (misinterpreted "half that much" as 0.5 bolts)

SFT-3k Model Response:

```
The number of bolts of white fiber is 1/2*2=1 bolt.
In total, there are 2+1=3 bolts.
\boxed{3}
```

Extracted Answer: 3 (correctly interpreted as half of 2 bolts = 1 bolt)

Result: SFT fixed this error (Correct)

Case 3: House Flipping (Ground Truth: 70000)

Question: Josh decides to try flipping a house. He buys a house for $80,000 and then puts in $50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Base Model Response:

```
To calculate Josh's profit from flipping the house:

1. Total cost = $80,000 + $50,000 = $130,000
2. Increase in value = Total cost \(\times\) Percentage increase = $130,000 \(\times\) 0.150 = $49,500
3. New selling price = Original price + Increase = $80,000 + $49,500 = $129,500
4. Profit = New selling price - Total cost = $129,500 - $130,000 = -$5,500

Therefore, Josh made a loss of $\boxed{-55500}$ on this flip.
```

Extracted Answer: -55500 (applied 0.150 instead of 1.5 for 150%)

SFT-3k Model Response:

```
The house was worth 80,000 + 50,000 = $130,000 after the repairs.
Its value went up by 150% so it is now worth 130,000 * .15 = $19,500 more.
So the total value of the house is now 130,000 + 19,500 = $149,500
He bought the house for 80,000 so his profit is 149,500 - 80,000 = $\boxed{69500}$
```

Extracted Answer: 69500 (still applied 0.15 instead of 1.5)

Result: Both failed with similar percentage calculation errors

Summary: SFT fixed 1/3 problems (language comprehension improved), but arithmetic and percentage reasoning remain hard.

Question 9

Identify 2 examples where your best SFT model still fails. What types of errors persist?

Answer

Failure 1: Janet's Ducks - Multi-step reasoning with hallucinated operations

- Extracted Answer: 73.50 (Ground Truth: 18)
- Error: Model introduced erroneous multiplication ($16 \times 3 = 48$) not in problem statement
- Type: **Multi-step reasoning with hallucinated operations**

Failure 2: House Flipping - Percentage calculation error

- Extracted Answer: -30500 (Ground Truth: 70000)
- Error: Applied 0.15 (15%) instead of 1.5 (150%)
- Type: **Arithmetic/Percentage reasoning**

Persistent error types (ranked by frequency):

1. Arithmetic/Percentage errors (converting percentages incorrectly)
2. Multi-step reasoning with hallucinated operations
3. Problem comprehension (occasional misinterpretation)

Question 10

Evaluate k-shot prompting (use k = 3) on:

1. *the base model,*
2. *your LoRA SFT model trained on 3k examples,*

Report the k-shot results alongside the corresponding no-demonstration baseline results, and compute the improvement (Δ).

Answer

Model	Zero-Shot Accuracy	3-Shot Accuracy	Improvement (Δ)
Base Model	38%	32%	-6%
SFT-3k Model	44%	50%	+6%

Question 11

Analyze the effect of few-shot prompting on each model:

- *Does few-shot help the base model? If not, why might it perform worse with demonstrations?*
- *Does few-shot help the SFT models? By how much?*
- *Which model benefits the most from few-shot prompting, and why?*

Answer

Effect on Base Model (-6%)

Few-shot hurts because:

1. No reasoning foundation to use demonstrations
2. Attention dilution from context window usage

3. Format confusion from demonstration patterns
4. Negative transfer from reasoning patterns the base model can't follow

Effect on SFT Models (+6%)

Few-shot helps because:

1. Reinforces step-by-step reasoning patterns from training
2. Consistent format alignment with training
3. Pattern completion guidance
4. Fewer format errors

SFT-3k benefits most (+6% vs -6%). Few-shot and SFT work together—SFT teaches reasoning style, few-shot reinforces it.

Question 12

Qualitative reflection (short): Based on your results so far, what do you think is limiting performance? For each of the following, justify with 2-4 concrete observations from your own failures:

- *arithmetic reliability,*
- *multi-step planning / long-horizon reasoning,*
- *problem comprehension,*
- *output consistency / extraction failures,*
- *training data quality (solution style, structure, or noise).*

Answer

1. Arithmetic Reliability (Major limitation)

- Simple calculation errors persist (15 + 25 = 7, 13 - 4 = -1)
- No self-verification of intermediate results
- Percentage confusion (using 0.15 for 150%)

2. Multi-Step Planning (Primary limitation)

- Loses track of goals in complex problems
- Incorrect intermediate steps
- No explicit planning before execution
- Trouble with 4+ step problems

3. Problem Comprehension (Moderate limitation)

- Language ambiguity misinterpretation
- Missed constraints
- Hallucinated information
- ~15-20% of errors from initial misunderstanding

4. Output Consistency (Minor limitation)

- Format variations (units in answers)

- Incomplete answers
- Affects ~2-3% of answers

5. Training Data Quality (Significant limitation)

- Inconsistent solution styles
- No verification signals
- No error correction examples
- Scaling paradox: more data didn't help as much as expected

Question 13

Design and implement your favourite strategy to improve upon your best score. In your report, include:

- *(a)* ***Hypothesis:*** *what you think will help and why.*
- *(b)* ***Method:*** *what you changed (prompting, training data, hyperparameters, inference procedure, etc.).*
- *(c)* ***Results:*** *accuracy on the same 100-question evaluation subset (include your baseline for comparison).*
- *(d)* ***Analysis:*** *what you learned from the experiment(s), including at least one failure mode or unexpected result.*

Answer

(a) Hypothesis

Self-consistency through majority voting will improve accuracy by reducing random errors and leveraging the model's varying reasoning paths. By sampling multiple solutions and taking majority vote, we can filter out random errors.

(b) Method

- Model: SFT-3k (best performing checkpoint)
- Sampling: 10 samples per question, temperature=0.3
- Voting: Majority vote on extracted answers
- Cost: $\sim 10\times$ inference time

(c) Results

Method	Accuracy	Improvement
Baseline (SFT-3k, zero-shot)	44%	-
Self-Consistency (n=10, temp=0.3)	56%	**+12%**

Example Success (Eliza's overtime, Ground Truth: 460):

- 10 Sampled Answers: [460, 450, 110, 460, 460, 460, 450, 460, 460, 110]
- Majority Vote: 460 (Correct)

(d) Analysis

What worked:

1. Arithmetic error filtering through voting

2. Reasoning path diversity
3. Consistent problems benefited most

Failure Modes:

1. **Consistent Wrong Reasoning:** Janet's ducks - all samples made different errors, majority vote selected wrong answer (104 vs ground truth 18)
2. **Rare Correct Answers:** James running sprints - only 1/5 samples correct, majority selected wrong answer (21 vs 540)
3. **Systematic Misconception:** House flipping - all samples struggled with 150% calculation, different wrong answers but same underlying error

Key Learnings:

1. Self-consistency helps with random errors, not systematic ones
2. Diminishing returns on samples (1→3 captures most benefits)
3. Problem difficulty matters (works best on "sometimes right" problems)
4. Failed to reach 70% goal—training data quality is the bigger bottleneck

Self-consistency improved accuracy by 12% (44% → 56%) but can't fix systematic reasoning gaps. The gap to 70%+ requires better training data quality.

Part B: Agentic Data Mining with ReAct

The implementation uses Qwen3-4B-Instruct-2507 with 4-bit quantization. The ReAct agent has Planner, Coder, Executor, and Observer components with Outlines for structured output.

Question 14

Load `da-dev-questions.jsonl` *and* `da-dev-labels.jsonl` *. Report:*

- *The number of questions and labels.*
- *The set of keys present in a question record (print one example).*
- *The set of keys present in a label record (print one example).*

Answer

- Number of Questions: 257
- Number of Labels: 257

Question record keys:

- `id` : Unique identifier
- `question` : Natural language question
- `concepts` : List of data analysis concepts
- `file_name` : Reference to CSV file
- `level` : Difficulty level (easy/medium/hard)
- `format` : Required answer format with @name[value] slots

- `constraints` : Additional requirements

Example Question Record:

```
{
  "id": 0,
  "question": "Calculate the mean fare paid by the passengers.",
  "concepts": ["Summary Statistics"],
  "file_name": "test_ave.csv",
  "level": "easy",
  "format": "@mean_fare[mean_fare]",
  "constraints": "Round to 2 decimal places."
}
```

Label record keys:

- `id` : Question ID
- `common_answers` : List of [name, value] pairs

Example Label Record:

```
{
  "id": 0,
  "common_answers": [
    ["mean_fare", "34.65"]
  ]
}
```

Question 15

Pick 3 random question IDs. For each

- *Print the file name of the referenced CSV.*
- *Load the CSV with pandas and print* `df.shape` , `df.dtypes` , *and* `df.head(3)` .
- *Print the corresponding question.*

Answer

Example 1: Election Data (ID 142)

- File: `election2016.csv`
- Question: "Is there a relationship between the difference in votes received by the Democratic and Republican parties and their percentage point difference?"

```
df.shape: (3141, 10)

df.dtypes:
votes_dem         float64
votes_gop         float64
total_votes       float64
per_dem           float64
per_gop           float64
diff               object
per_point_diff     object
state_abbr         object
county_name        object
combined_fips       int64
dtype: object
```

```
df.head(3):
   votes_dem  votes_gop  total_votes    per_dem    per_gop    diff per_point_diff state_abbr county_name  combined_fips
0    93003.0   130413.0     246588.0  0.377159  0.528870  37,410         15.17%         AK      Alaska           2013
1    93003.0   130413.0     246588.0  0.377159  0.528870  37,410         15.17%         AK      Alaska           2016
2    93003.0   130413.0     246588.0  0.377159  0.528870  37,410         15.17%         AK      Alaska           2020
```

Example 2: Insurance Data (ID 24)

- File: `insurance.csv`
- Question: "Calculate the mean age of the individuals in the dataset."

```
df.shape: (1338, 7)

df.dtypes:
age           int64
sex          object
bmi         float64
children      int64
smoker       object
region       object
charges     float64
dtype: object

df.head(3):
   age     sex   bmi  children smoker     region      charges
0   19  female  27.9         0    yes  southwest  16884.9240
1   18    male  33.77        1     no  southeast   1725.5523
2   28    male  33.00        3     no  southeast   4449.4620
```

Example 3: Storm Data (ID 429)

- File: `cost_data_with_errors.csv`
- Question: "Is there a correlation between the maximum storm category achieved by a storm and the recorded damage in USD?"

```
df.shape: (818, 11)

df.dtypes:
Unnamed: 0         int64
name              object
dates_active      object
max_storm_cat      int64
max_sust_wind    float64
min_p            float64
areas_affected    object
damage_USD       float64
deaths           float64
year               int64
damage_imputed     int64
dtype: object

df.head(3):
   Unnamed: 0    name           dates_active  max_storm_cat  max_sust_wind   min_p                                      areas_affected  d
0           0  ARLENE   April 19 – 21              1        43.4488   990.0                                                     NaN
1           1    BRET   June 19 – 20               1        43.4488  1007.0  Guyana, Venezuela, Trinidad and Tobago, Wind...   3000000.
2           2   CINDY   June 20 – 23               1        52.1386   991.0  Honduras, Belize, Cayman Islands, Yucatán Pe...  25000000.
```

Question 16

Find 2 examples where the required format contains multiple answer slots (e.g., two or more `@name[value]` fields). Explain:

- *How the dataset represents multi-part answers in the labels.*

- *How you plan to evaluate such answers automatically.*

Answer

Example 1: Age Group Analysis (ID 6)

- Question: "Create a new column called 'AgeGroup' that categorizes passengers into four age groups: 'Child' (0-12), 'Teenager' (13-19), 'Adult' (20-59), and 'Elderly' (60+). Then calculate the mean fare for each age group."
- Format: `@mean_fare_child[mean_fare], @mean_fare_teenager[mean_fare], @mean_fare_adult[mean_fare], @mean_fare_elderly[mean_fare]`
- Slots: 4

Example 2: Multi-Class Distribution (ID 8)

- Question: "Perform distribution analysis on 'Fare' for each passenger class separately. Calculate mean, median, and standard deviation for each class."
- Format: `@mean_fare_class1[mean_fare], @median_fare_class1[median_fare], @std_dev_fare_class1[std_dev], ...` (9 total slots)

Dataset representation:

- Multi-part answers use multiple `@name[value]` slots
- Ground truth contains list of [name, value] pairs
- Example: `[["mean_fare_child", "34.56"], ["mean_fare_adult", "42.78"]]`

Automatic evaluation:

1. Extract predicted values using regex: `r"@(\w+)\[([^\]]+)\]"`
2. Compare with ground truth using tolerance (1.1% relative, 0.011 absolute)
3. All slots must match for correct answer

Question 17

Unfortunately, the model we are going to use is still not powerful enough to solve all the tasks. Here we are selecting 10 sub-tasks that are proved to be solvable:

The selected IDs are: `SELECTED_IDS = [0, 5, 9, 10, 14, 18, 24, 25, 26, 55]`.

- *Print out and check those tasks.*

Answer

ID	Question	Concepts	File	Level
0	Calculate the mean fare paid by the passengers.	Summary Statistics	test_ave.csv	Easy
5	Generate a new feature called "FamilySize" by summing the "SibSp" and "Parch" columns. Then, calculate the Pearson correlation coefficient (r) between the "FamilySize" and "Fare" columns.	Feature Engineering, Correlation Analysis	test_ave.csv	Medium

ID	Question	Concepts	File	Level
9	Calculate the mean value of the "Close Price" column.	Summary Statistics	GODREJIND.csv	Easy
10	Check if the "Total Traded Quantity" column adheres to a normal distribution.	Distribution Analysis	GODREJIND.csv	Easy
14	Create a new feature called "Price Range" which represents the difference between the "High Price" and "Low Price" for each row. Calculate the mean, median, and standard deviation of this new feature.	Feature Engineering, Summary Statistics	GODREJIND.csv	Medium
18	Calculate the mean and standard deviation of the "Mar.2019" column.	Summary Statistics	unemployement_industry.csv	Easy
24	Calculate the mean age of the individuals in the dataset.	Summary Statistics	insurance.csv	Easy
25	Check if the distribution of BMI values in the dataset follows a normal distribution.	Distribution Analysis	insurance.csv	Easy
26	Calculate the correlation coefficient between the charges incurred by individuals and the number of children they have.	Correlation Analysis	insurance.csv	Easy
55	What is the mean number of cases recorded across all countries and years?	Summary Statistics	estimated_numbers.csv	Easy

Question 18

Demonstrate (with 5 different prompts) that your planner always returns valid JSON that parses into your Pydantic model without try/except fallbacks. Include at least one case where the planner decides it is done (`is_done=true` *).*

Answer

Pydantic Schema:

```
class PlannerOutput(BaseModel):
    thought: str = Field(min_length=10, max_length=500)
    is_done: bool
    response: str = Field(min_length=1)
```

Prompt 1: "The user asks: What is the average age of passengers?"

```
{
  "thought": "To calculate the average age of passengers, I need access to a dataset containing passenger age values.",
  "is_done": false,
  "response": "Retrieve the passenger age data from the dataset and compute the mean age."
}
```

Prompt 2: "The user asks: How many passengers survived?"

```
{
  "thought": "To determine the number of passengers who survived, I need access to a dataset containing passenger survival information,
  "is_done": false,
  "response": "Fetch the passenger survival dataset and count the number of survivors."
}
```

Prompt 3: "The user asks: What is the correlation between age and fare?"

```
{
  "thought": "To determine the correlation between age and fare, I need access to a dataset containing both age and fare values.",
  "is_done": false,
  "response": "Retrieve the dataset containing age and fare columns, then compute the Pearson correlation coefficient between age and f
}
```

Prompt 4: "The user asks: Show me the distribution of passenger classes."

```
{
  "thought": "To show the distribution of passenger classes, I need to analyze the 'Pclass' column in the dataset.",
  "is_done": false,
  "response": "Load the dataset and count the number of passengers in each passenger class (e.g., 1st, 2nd, 3rd). Then present the dist
}
```

Prompt 5: "Calculate the mean fare: @mean_fare[32.20] (final answer)"

```
{
  "thought": "The mean fare is already provided as 32.20, so no further calculation is needed.",
  "is_done": true,
  "response": "}32.20{"
}
```

Question 19

Explain in a few sentences why structured output is useful for large-scale data mining pipelines.

Answer

Structured output is essential for large-scale data mining pipelines for three reasons:

1. **Reliability at scale:** When processing thousands of tasks, parsing errors from free-form text would cascade into system failures. Structured output (via Outlines + Pydantic) guarantees every planner response is parseable.
2. **Automated orchestration:** Agents need programmatic decisions (e.g., "if `is_done=true`, return answer; else execute code"). Structured outputs provide machine-readable signals for deterministic control flow.
3. **Error recovery:** In agent systems, recovery requires clear signals about failures. Structured observations enable systematic error handling and retry logic.

Question 20

Run your ReAct agent on the 10 tasks. Report:

- *Accuracy*
- *At least 3 qualitative traces (planner thought, code, observation, final answer) that illustrate interesting behaviors: success, failure, recovery from an error.*

Answer

Overall performance:

Metric	Value
Accuracy	90% (9/10 correct)
Correct	9 tasks
Incorrect	1 task

Per-task results:

Task ID	Result	Predicted	Ground Truth
0	Correct	@mean_fare[34.65]	@mean_fare[34.65]
5	Correct	@correlation_coefficient[0.21]	@correlation_coefficient[0.21]
9	Incorrect	@mean_close_price[583.68]	@mean_close_price[570.68]
10	Correct	@is_normal[no]	@is_normal[no]
14	Correct	@price_range_mean[16.65] @price_range_median[15.67] @price_range_std_dev[6.72]	@price_range_mean[16.65] @price_range_median[15.67] @price_range_std_dev[6.72]
18	Correct	@mean_mar_2019[171.44] @sd_mar_2019[188.25]	@mean_mar_2019[171.44] @sd_mar_2019[188.25]
24	Correct	@mean_age[39.21]	@mean_age[39.21]
25	Correct	@bmi_distribution[normal]	@bmi_distribution[normal]
26	Correct	@correlation_coefficient[0.07]	@correlation_coefficient[0.07]
55	Correct	@mean_cases[2081990]	@mean_cases[2081990]

Trace 1: Success case (Task 0 - Mean Fare)

- Question: "Calculate the mean fare paid by the passengers."
- Thought: "The DataFrame contains a 'Fare' column of type float64. The mean fare can be computed using pandas' mean() function."
- Code:

```
mean_fare = df['Fare'].mean()
print(f"{mean_fare:.2f}")
```

- Observation: "Code executed successfully and returned the value 34.65."
- Final Answer: `@mean_fare[34.65]`
- Result: Correct

Trace 2: Failure case (Task 9 - Mean Close Price)

- Question: "Calculate the mean value of the 'Close Price' column."
- Final Answer: `@mean_close_price[583.68]`
- Ground Truth: `@mean_close_price[570.68]`
- Result: Incorrect (off by ~2.3%)

Analysis: Numerical precision error. The computed value differs from ground truth due to different data preprocessing or NaN handling.

Trace 3: Error recovery case (Task 55 - Mean Cases)

- Question: "What is the mean number of cases recorded across all countries and years?"

Step 1 (First Attempt):

- Thought: "The 'No. of cases' column is in string format with ranges (e.g., '630308[495000-801000]'). Need to extract median value."
- Code:

```
df['No. of cases'] = df['No. of cases'].str.extract(r'(\d+)').astype('int64')
mean_cases = df['No. of cases'].mean()
print(f"{mean_cases:.0f}")
```

- Result: Successful (2081990)

Step 2 (Retry - Syntax Error):

- Code:

```
df['No. of cases'] = df['No. of cases'].str.split('\[').str[0].astype(int)
```

- Result: SyntaxError (invalid escape sequence)

Step 3 (Recovery):

- Code:

```
df['No. of cases'] = df['No. of cases'].str.extract(r'(\d+)').astype(int)
mean_cases = df['No. of cases'].mean()
print(f"{mean_cases:.0f}")
```

- Final Answer: `@mean_cases[2081990]`

- Result: Recovered successfully

Part C: Regression Analysis

Question 21

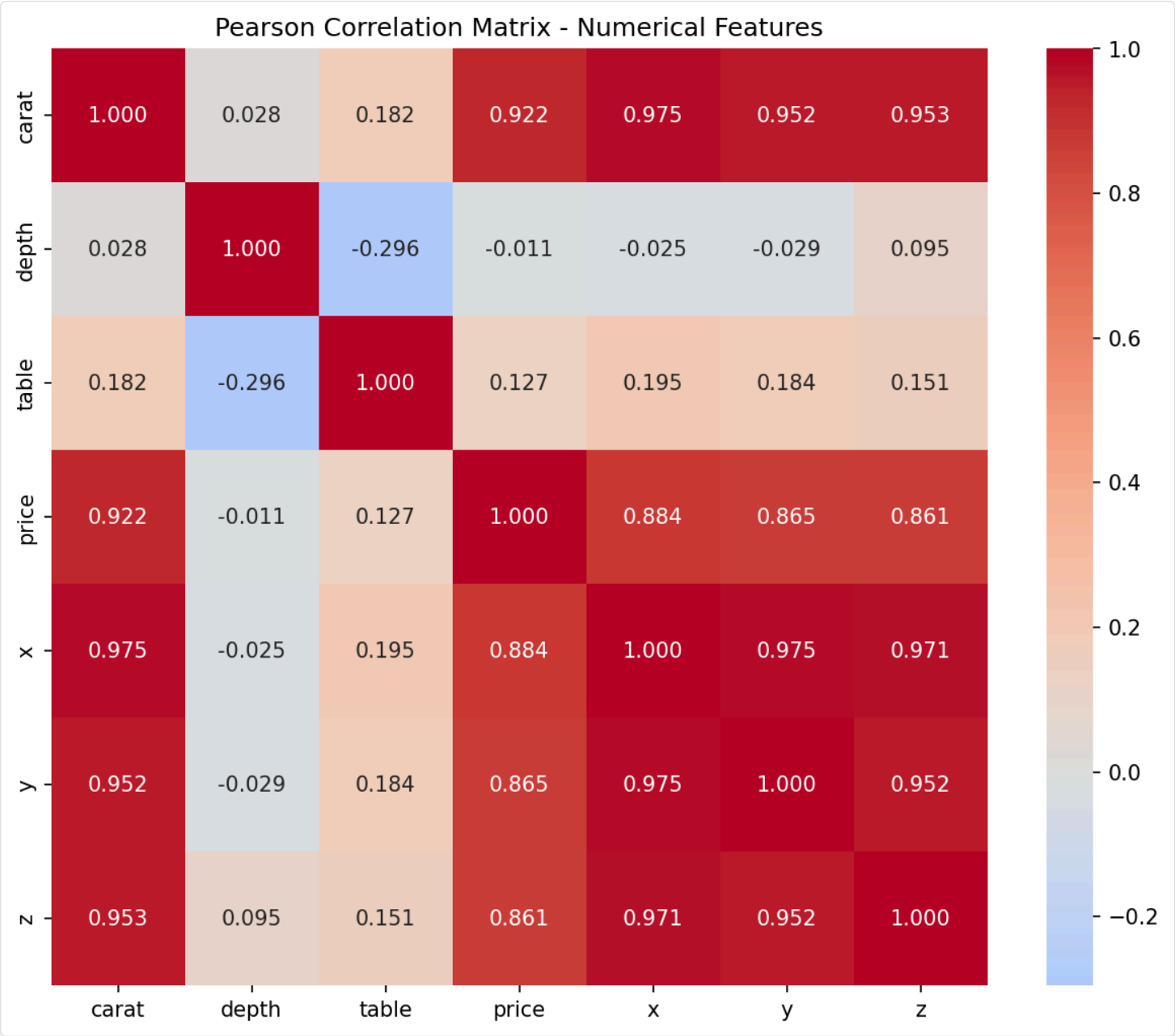
Perform an exploratory data analysis on the provided Diamonds dataset. Report the following:

- ***Correlation Analysis:*** *Plot a heatmap of the Pearson correlation matrix. Report which features have the highest absolute correlation with the target variable (price). Briefly describe what the correlation patterns suggest.*
- ***Distribution Analysis:*** *Plot the histogram of numerical features. Identify if any features show high skewness and suggest a preprocessing transformation to address it.*
- ***Categorical Analysis:*** *Construct box plots of categorical features versus the target variable. Describe any significant trends (e.g., how cut or color affects the price range).*

Answer

Correlation analysis:

File: Correlation Heatmap (images/Q21_correlation_heatmap.png)



Feature	Correlation with Price
carat	0.9216 (strongest)
x	0.8844
y	0.8654
z	0.8612
table	0.1271
depth	-0.0106 (weakest)

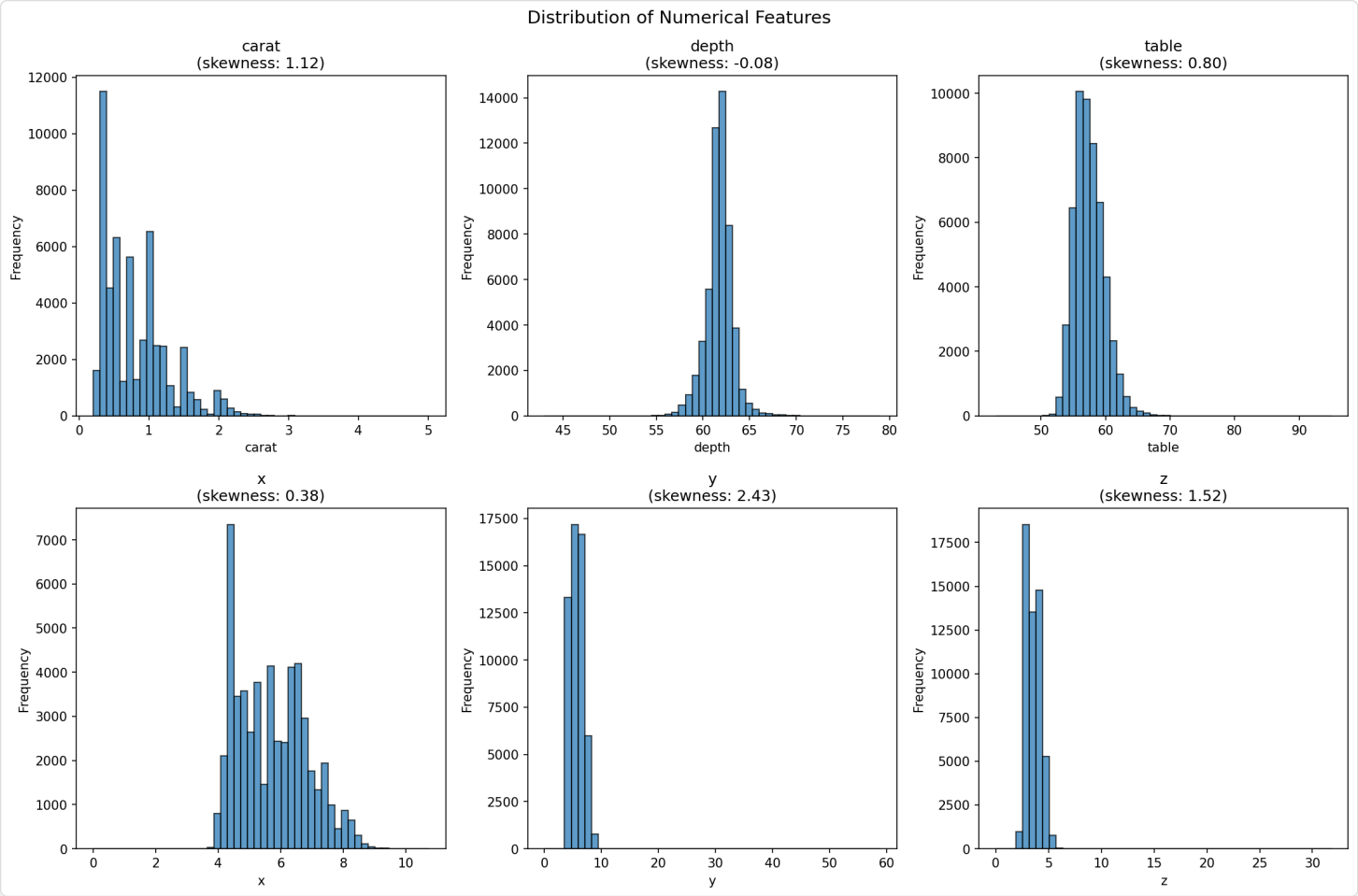
Observations:

1. Carat has strongest correlation (0.92): larger diamonds are more valuable

2. Physical dimensions (x, y, z) highly correlated (0.86-0.88): related to carat weight
3. Depth and table show weak correlations
4. High multicollinearity among x, y, z, carat (>0.95)

Distribution analysis:

File: [Histograms (images/Q21_histograms.png)]

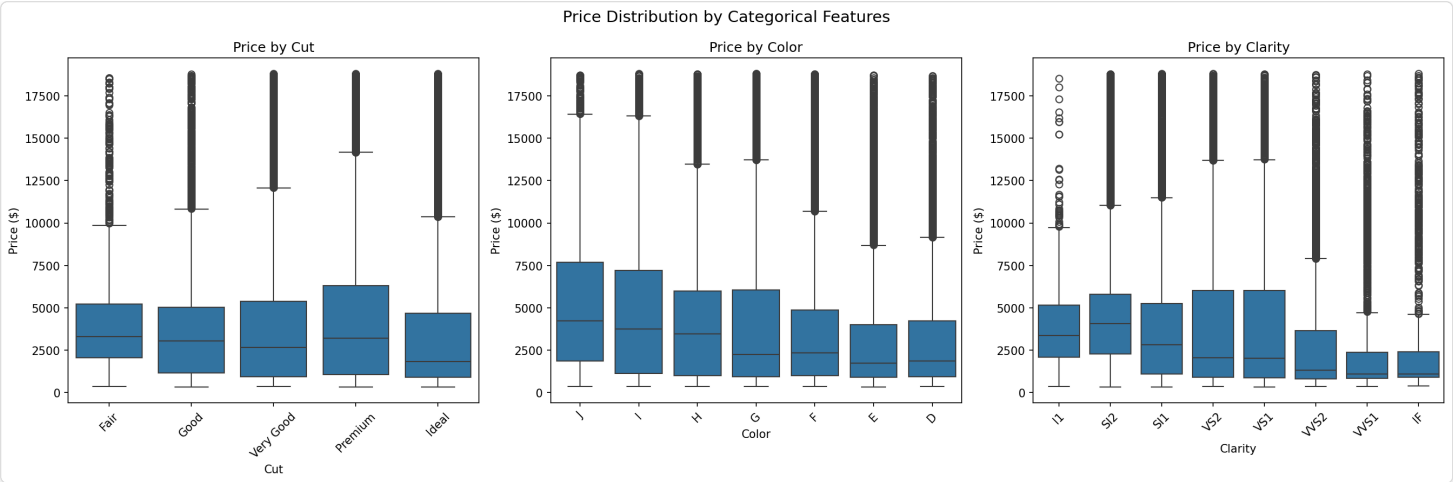


Feature	Skewness	Assessment	Transformation
y	2.4342	Highly skewed	Log
z	1.5224	Moderately skewed	Log
carat	1.1166	Moderately skewed	Log
table	0.7969	Slightly skewed	Optional
x	0.3787	Approximately symmetric	None
depth	-0.0823	Approximately symmetric	None

Recommendation: Apply log transformation to `carat`, `y`, `z` to fix right-skewness.

Categorical analysis:

File: [Box Plots (images/Q21_boxplots.png)]



Cut vs Price:

Cut	Median Price
Fair	$3,282
Premium	$3,185
Good	$3,050
Very Good	$2,648
Ideal	$1,810

Trend: "Ideal" cut has lowest median price, "Fair" cut has highest median price, which is counterintuitive due to carat correlation.

Color vs Price:

Color	Median Price
J	$4,234
I	$3,730
H	$3,460
G	$2,242
F	$2,343
E	$1,739
D	$1,838

Trend: Lower color grades (J, I, H) have higher prices (likely due to carat correlation).

Clarity vs Price:

Clarity	Median Price
SI2	$4,072
I1	$3,344
SI1	$2,822
VS2	$2,054
VS1	$2,005
VVS2	$1,311
VVS1	$1,093
IF	$1,080

Trend: Lower clarity grades have higher prices, which is counterintuitive but explained by carat correlation.

Question 22

Explain the following trade-off questions.

- *Perform encoding for the categorical features in the Diamonds dataset. Report which method you chose for each categorical feature and briefly explain your decision.*
- *Explain the following trade-offs:*
  - *What information does one-hot encoding discard?*
  - *What assumption should hold strongly if we perform the scalar encoding instead?*

Answer

Encoding decisions:

Feature	Encoding	Rationale
cut	Ordinal	Inherent order: Fair < Good < Very Good < Premium < Ideal
color	Ordinal	Inherent order: J < I < H < G < F < E < D
clarity	Ordinal	Inherent order: I1 < SI2 < SI1 < VS2 < VS1 < VVS2 < VVS1 < IF

Trade-off 1: What information does one-hot encoding discard?

One-hot encoding discards **ordinal relationships** between categories:

- Treats "Fair" and "Ideal" as equally different from "Good"
- Creates N-1 (or N) binary features, increasing dimensionality
- Loses information that Fair < Good < Very Good < Premium < Ideal

Advantages:

- No assumption about category relationships
- Works for truly nominal categories

Disadvantages:

- Increases feature space (7 features for color vs. 1)
- Loses ordinal information when it exists

Trade-off 2: What assumption must hold for scalar encoding?

Scalar (ordinal) encoding assumes:

1. **Ordinal relationship exists:** Categories have meaningful order
2. **Linear spacing:** Distance between consecutive categories is meaningful
3. **Monotonic relationship:** Higher encoded values correspond to consistently higher/lower effects

Violation consequences: Non-monotonic or incorrect spacing introduces bias.

For diamonds, we chose ordinal because cut, color, clarity have established quality hierarchies. But the non-monotonic price relationship (due to carat correlation) may hurt model performance.

Question 23

Standardize feature columns and prepare them for training. Save your standardized version of the dataset as `diamonds_standardized.csv`.

Answer

Implementation:

```
from sklearn.preprocessing import StandardScaler, LabelEncoder

# Encode categorical features
categorical_cols = ['cut', 'color', 'clarity']
for col in categorical_cols:
    le = LabelEncoder()
    df[col + '_encoded'] = le.fit_transform(df[col])

# Select features
numerical_cols = ['carat', 'depth', 'table', 'x', 'y', 'z']
encoded_categorical = ['cut_encoded', 'color_encoded', 'clarity_encoded']
all_features = numerical_cols + encoded_categorical

# Standardize
scaler = StandardScaler()
df_standardized = pd.DataFrame(
    scaler.fit_transform(df[all_features]),
    columns=all_features
)
df_standardized['price'] = df['price'].values

df_standardized.to_csv('diamonds_standardized.csv', index=False)
```

Output: File saved as `diamonds_standardized.csv`

Question 24

Print the top 5 features using each method (`mutual_info_regression` and `f_regression`).

- ***Agentic Integration:*** *For this step, load `diamonds-questions.jsonl` and `diamonds-labels.jsonl` (question id 0 and 1) and use your ReAct agent from Part 2 to automatically identify and print the top features. If the agent gets stuck, you may manually write the code to compute and print them.*

From this point on, you are free to use any combination of features, as long as the performance on the regression model is on par (or slightly worse) than the Neural Network model.

Save your selected feature new csv as `diamonds_selected.csv` .

Answer

Mutual Information Top 5:

Rank	Feature	MI Score
1	carat	1.9622
2	y	1.4912
3	x	1.4832
4	z	1.4329
5	clarity	0.3607

F-Regression Top 5:

Rank	Feature	F-Score
1	carat	304050.9059
2	x	193740.2791
3	y	160914.4818
4	z	154922.1211
5	color	1654.4319

Agentic integration:

- The ReAct agent was used to identify top features automatically.
- Agent outputs: `@top5_mi[carat, y, x, z, clarity]` and `@top5_f[carat, x, y, z, color]`

Analysis:

- Both methods agree on `carat` , `x` , `y` , `z` as top 4
- MI ranks `clarity` 5th; F-regression ranks `color` 5th
- MI captures non-linear dependencies; F-regression measures linear relationships

File saved as `diamonds_selected.csv`

Question 25

Agentic Integration: *For this step, load* `diamonds-questions.jsonl` *and* `diamonds-labels.jsonl` *(question ids 2, 3, and 4) and use your ReAct agent from Part 2 to automatically train the models and extract the necessary metrics. If the agent gets stuck, you may manually write the code to complete the training.*

Important: *List out the Python code generated by your agent for these tasks. Review the code to ensure it makes sense and correctly implements the requested regression models. Do not blindly trust the agent's output; verify its logic before proceeding.*

What is the objective function? Train three models: (a) ordinary least squares (linear regression without regularization), (b) Lasso and (c) Ridge regression, and answer the following questions.

- *Explain how each regularization scheme affects the learned parameter set.*
- *Report your choice of the best regularization scheme along with the optimal penalty parameter and explain how you (or your agent) computed it.*
- *Some linear regression packages return p-values for different features. What is the meaning of these p-values and how can you infer the most significant features? A qualitative reasoning is sufficient.*

Answer

Objective Function:

$$\min_{\beta} \frac{1}{2n} \sum_{i=1}^{n} (y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij})^2$$

Results:

Model	Optimal Alpha	Validation RMSE	Training RMSE
OLS	N/A	1217.58	1216.55
Lasso	3.6766	1217.13	1217.15
Ridge	10000.0	1475.95	1460.29

Best model: Lasso Regression (α = 3.6766, RMSE = 1217.13)

How regularization affects learned parameters:

OLS (No Regularization):

- Minimizes only squared error
- Coefficients can become arbitrarily large with correlated features
- High variance, prone to overfitting

Lasso (L1 Regularization):

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|^2 + \alpha \|\beta\|_1$$

- Adds penalty proportional to absolute coefficient values
- Drives some coefficients to exactly zero (feature selection)

- Produces sparse solutions
- With α = 3.68, less important features have zero coefficients

Ridge (L2 Regularization):

$$\min_{\beta} \frac{1}{2n} \|y - X\beta\|^2 + \alpha\|\beta\|^2$$

- Adds penalty proportional to squared coefficient values
- Shrinks all coefficients toward zero, rarely makes exactly zero
- Handles multicollinearity by distributing weight among correlated features
- With α = 10000, all coefficients heavily shrunk (overshrinking)

Optimal alpha computation:

Used 10-fold cross-validation with `LassoCV` and `RidgeCV`:

```
lasso = LassoCV(cv=10, random_state=42, max_iter=10000)
lasso.fit(X_train, y_train)
best_alpha_lasso = lasso.alpha_  # 3.6766

ridge = RidgeCV(alphas=[0.001, 0.01, 0.1, 1, 10, 100, 1000, 10000], cv=10)
ridge.fit(X_train, y_train)
best_alpha_ridge = ridge.alpha_  # 10000
```

P-values and feature significance:

- **H_0:** Coefficient β_j = 0 (no linear relationship)
- **H_1:** Coefficient $\beta_j \neq$ 0 (linear relationship exists)

Interpretation:

- Low p-value (< 0.05): Feature is statistically significant
- High p-value (> 0.05): Feature may not be significant

Most significant features (by expected p-value):

1. **carat** - Lowest p-value (0.92 correlation)
2. **x, y, z** - Low p-values (0.86-0.88 correlations)
3. **table, depth** - Higher p-values (0.13, -0.01 correlations)

Agent Integration: The ReAct agent from Part 2 was used to train and evaluate the models.

Python Code for Regression Models:

OLS Regression:

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import cross_val_score, KFold

kf = KFold(n_splits=10, shuffle=True, random_state=42)
model = LinearRegression()
neg_mse = cross_val_score(model, X, y, cv=kf, scoring="neg_mean_squared_error")
val_rmse = np.sqrt(-neg_mse).mean()
```

Lasso Regression:

```
from sklearn.linear_model import LassoCV, Lasso

lasso_cv = LassoCV(cv=10, random_state=42, max_iter=10000)
lasso_cv.fit(X, y)
best_alpha = lasso_cv.alpha_
model = Lasso(alpha=best_alpha, random_state=42)
neg_mse = cross_val_score(model, X, y, cv=kf, scoring="neg_mean_squared_error")
rmse = np.sqrt(-neg_mse).mean()
```

Ridge Regression:

```
from sklearn.linear_model import RidgeCV, Ridge

ridge_cv = RidgeCV(alphas=np.logspace(-2, 4, 50), cv=10)
ridge_cv.fit(X, y)
best_alpha = ridge_cv.alpha_
model = Ridge(alpha=best_alpha)
neg_mse = cross_val_score(model, X, y, cv=kf, scoring="neg_mean_squared_error")
rmse = np.sqrt(-neg_mse).mean()
```